

TITLE

Advancing Harmless AI Assistants Through Constitutional AI and Reinforcement Learning

ABSTRACT

This research investigates the efficacy of Constitutional AI (CAI) and Reinforcement Learning from AI Feedback (RLAIF) in developing harmless yet helpful AI assistants. We present findings demonstrating that CAI can train models preferred over those trained with human feedback, achieving high harmlessness without explicit human labels for harmfulness. The study details methods like chain-of-thought reasoning and critique-revision pipelines, showing their impact on reducing AI-generated harms. The results indicate that RLAIF significantly enhances harmlessness with minimal impact on helpfulness, suggesting a scalable and effective approach for building safer AI.

1. Introduction

The development of safe and beneficial artificial intelligence is a paramount concern. Large language models (LLMs) have demonstrated remarkable capabilities, but also exhibit propensity for generating harmful content. Traditional methods for mitigating these harms, such as Reinforcement Learning from Human Feedback (RLHF), are resource-intensive and rely on extensive human labeling. This paper explores Constitutional AI (CAI), a novel framework that leverages AI feedback (RLAIF) to train AI assistants to be harmless and helpful, potentially overcoming the limitations of human supervision. We investigate the effectiveness of CAI, particularly its critique and revision pipelines, and its

comparison against human-feedback based methods across various datasets and evaluation metrics.

2. Methods

2.1 Datasets

Our research utilizes a diverse array of datasets to evaluate AI assistant performance. These include: human-written prompts for red teaming (42,496), model-generated prompts by few-shot prompting a pre-trained model (140,335), human-written helpfulness prompts (135,296), human feedback data for preference model comparisons (collected in prior work), helpfulness and harmlessness data collected separately by crowdworkers, a dataset from [Ganguli et al., 2022] for harm-focused multiple choice evaluations, PALMS prompts, InstructGPT prompts, LaMDA prompts, and survey data from 399 undergraduate and postgraduate students from various disciplines in Hong Kong.

2.2 Algorithms

We employ several key algorithmic approaches. Constitutional AI (CAI) serves as the primary framework. Within CAI, we utilize a supervised learning phase (SL) and a reinforcement learning phase (RL). Specifically, the RL phase is driven by Reinforcement Learning from AI Feedback (RLAIF), contrasting with Reinforcement Learning from Human Feedback (RLHF). Other methods investigated include Scaling Supervision, Chain-of-thought (CoT) reasoning, a Critique and Revision pipeline, Self-improvement through AI feedback, and Preference model training.

2.3 Analytical Approach

The evaluation methodology involves fine-tuning a pretrained language model on revised responses using supervised learning. Subsequently, a preference model is trained from AI preferences. This preference model then serves as a reward signal for reinforcement learning, employing the RLAIIF approach. We also explore generating self-critiques and revisions, and an alternative method of direct revision generation. Ensembling over constitutional principles and clamping probabilities for CoT labels are also part of the analytical strategy. For comparison, models trained via RLHF and basic supervised learning (SL-CAI) are assessed.

3. Results

3.1 Key Findings

Constitutional AI (CAI) successfully trains a harmless but non-evasive AI assistant without requiring human labels for harmfulness. The AI feedback (RLAIIF) approach demonstrates the ability to train models preferred by crowdworkers over models trained with human feedback for harmlessness. Chain-of-thought (CoT) reasoning enhances AI's capacity to identify harms, leading to evaluations competitive with human feedback-trained preference models. Model-generated critiques and revisions progressively reduce harmfulness, with critiques proving more effective than direct revisions. Self-supervised preference labels for RL further improve model behavior, achieving or exceeding human feedback performance for harmlessness. Increasing the number of principles in the constitution improves response diversity without significantly boosting harmlessness scores. Critiqued revisions generally exhibit higher harmlessness than direct revisions, particularly for smaller models. RLAIIF models are

substantially more harmless than RLHF and SL-CAI models, with minimal compromise to helpfulness. RL-CAI models are highly effective at avoiding evasion and often provide nuanced, harmless responses to red team prompts.

3.2 Cross-Paper Comparison

Our findings from RLAIF-trained models indicate a significant preference from crowdworkers compared to RLHF models for harmlessness, suggesting a scalable advantage. The performance of CAI in generating harmless yet helpful responses aligns with the goals of other safety-focused research, but distinguishes itself by automating the labeling process. The effectiveness of critique-revision pipelines echoes findings in iterative refinement research, but the application to AI self-correction is novel. The minimal cost to helpfulness observed with RLAIF is a critical distinction from some prior RLHF studies which might see a more pronounced trade-off.

4. Conclusion

This research establishes Constitutional AI, particularly through Reinforcement Learning from AI Feedback (RLAIF), as a powerful and scalable paradigm for developing harmless and helpful AI assistants. The ability to train models that are preferred over those trained with human feedback, without explicit human labeling of harmfulness, represents a significant advancement. Techniques such as chain-of-thought reasoning and critique-revision pipelines further enhance AI safety by improving harm detection and mitigation. The results underscore that RLAIF offers a compelling alternative to RLHF, delivering superior harmlessness with minimal impact on helpfulness, paving the way for safer and more reliable AI deployments.

5. References

[Ganguli et al., 2022] (Specific citation details are not provided in the input and would be added here in a full research paper).